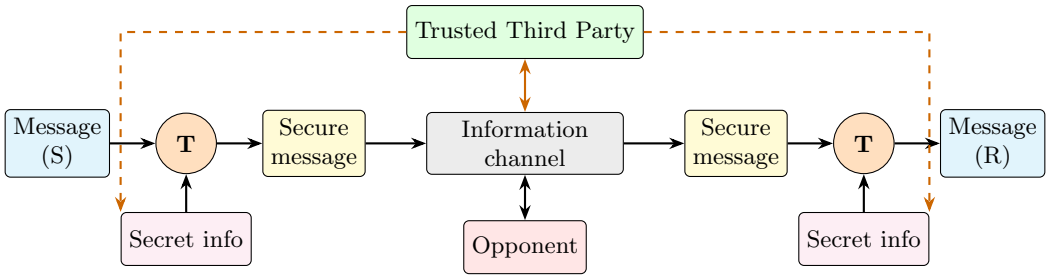
Cryptography & Network Security

Intro: Security Models, Cryptography Basics

1 Model for Network Security

Sender S wants to send a message to receiver R over an insecure channel (the Internet). An **opponent** can sniff the channel.



- **T (security-related transformation)** takes two inputs: the message and **secret info** (a key). Output: a secure message (ciphertext). R applies the reverse T with its secret info.
- It is called T, not E, because encryption is only one option. **Steganography** (hiding data inside an image, audio, etc.) is another.
- The opponent can intercept the secure message but cannot read it.
- Security rests on the **key** (its secrecy and bit length) and algorithm strength, not on hiding the algorithm. Small key space ⇒ brute force works.
- **Trusted Third Party (TTP)**: distributes keys, issues certificates, settles disputes. Example: a Certificate Authority signing a bank's SSL certificate. (Details come later in the course.)

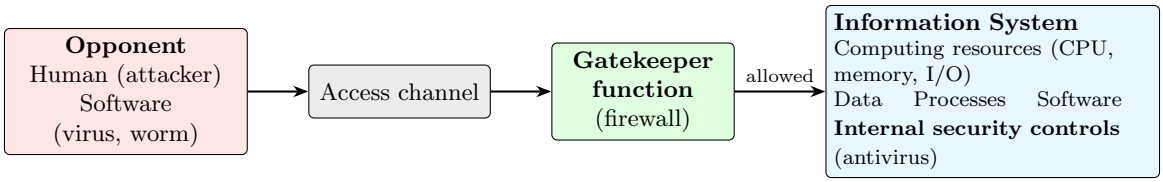
1.1 Four tasks in designing a security service

1. **Design an algorithm** for T (encrypt at S, decrypt at R) that the opponent cannot defeat.
2. **Generate the secret information** (large keys, so brute force is infeasible).
3. **Develop methods to distribute and share** the secret info (get keys to S and R safely).
4. **Specify a protocol**: the rules both parties follow to use the algorithm and keys (e.g., SSL/TLS, IPsec).

Memory aid: *Algorithm* → *Key* → *Distribute* → *Protocol*.

2 Network Access Security Model

Protects a *system* from unwanted access, rather than protecting a message in transit.



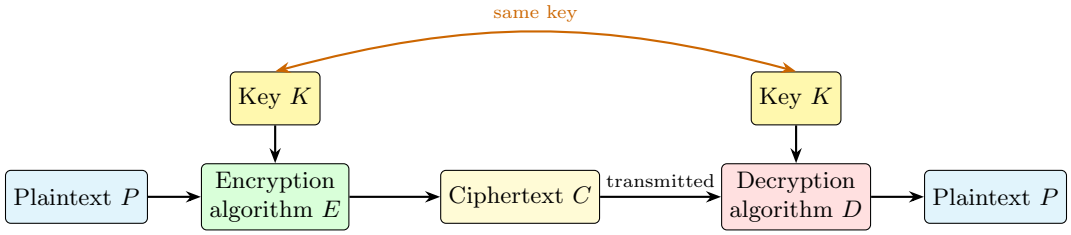
- **Opponent**: a human hacker or malicious software.
- **Gatekeeper**: first line of defence, like a guard at a gate. Filters traffic and blocks intruders (firewall, login checks).
- **Internal security controls**: second line, inside the system, catching what slips past the gate (antivirus, OS permissions).
- Insiders (e.g., employees) can also attack, but this model focuses on outside threats.

3 Cryptography

Definition

The art or science of transforming an **intelligible** message (plaintext) into an **unintelligible** one (ciphertext), and then **retransforming** it back to the original form.

3.1 Symmetric cipher model



$$C = E(K, P) \qquad P = D(K, C) = D\big(K, E(K, P)\big)$$

- Encryption needs **two** inputs (P and K); decryption needs C and K.
- E and D are **public**. Only K is secret (Kerckhoffs's principle).
- If K leaks, *all* past and future messages between S and R can be read.
- Toy example: $K = 100$ at both ends. Real keys are huge (128, 256, 2048 bits), so guessing is infeasible.

3.2 Types of cryptography

	Symmetric (private / secret key)	**Asymmetric** (public key)
Keys	One shared key	Key pair: public + private
Use	Same key encrypts and decrypts	One key encrypts, the other (mathematically related) decrypts
Secrecy	Key known only to S and R	Public key published; private key kept by owner
Toy example	Encrypt 100, decrypt 100	Encrypt 100, decrypt 500

Asymmetric works because the two keys satisfy a mathematical relationship, and the private key cannot practically be computed from the public one. (Which key does what is covered in a later lecture.)

4 Security of Encryption Schemes

Unconditionally secure

The ciphertext does **not contain enough information to uniquely determine the plaintext**, no matter how much ciphertext is available. Holds even against infinite computing power. Example: one-time pad.

Computationally secure (either condition)

1. **Cost** of breaking the cipher > **value** of the encrypted information.
2. **Time** to break the cipher > **useful lifetime** of the information.

- Cost: spending \$1,000,000 to steal a \$1,000 message is pointless.
- Time: a quote valid for 10 minutes that takes 50 years to crack is useless when cracked.
- Real ciphers (AES, RSA) are computationally secure; unconditional security needs a key as long as the message, which is impractical.

Exam tips

- Draw the network security model with all labels: S, R, T, secret info, secure message, opponent, TTP.
- Know the four tasks in order.
- "Why T and not E?" $\Rightarrow$ steganography is also a transformation.
- Symmetric = same key; asymmetric = different but related keys.
- Quote both conditions of computational security word for word.